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ODDO BHF

Team 6 White Paper

Re-Imagining the Future of Private Investment:
Jarvis - The AI Co-Manager That Beats the Market

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0. Key Terms

1) UUIDs: A standardized, 36-character alphanumeric label used to provide a unique identity to a specific piece of information so it can be recognized globally without the risk of duplication.

1. Executive Summary

Over the past decade, average equity fund fees in both Europe and the US have declined materially, while cost bases have continued to rise, compressing operating margins for active managers (Broadridge, 2024).

At the same time, competitive advantage increasingly depends on whether a manager can identify and act on information asymmetries earlier than the market. As McKinsey (2025a) notes, alpha generation is shifting from being primarily about access to information to being more about the ability to process complex, unstructured, and forward-looking data faster and more systematically than peers.

Jarvis is architected as a real-time alternative data intelligence layer that gathers unstructured alternative data, cleans it, interprets it, transforms it into concrete metrics for fundamental valuation workflows, and eventually provides PMs with market signals to construct a winning portfolio. The unconventional datasets include news flows, social and stakeholder sentiment, regulatory disclosures, land and asset registries, logistics and transportation data, and other real-world activity indicators (Hall, Jaffe & Trajtenberg, 2004).

Jarvis addresses two measurable value drivers for PMs:

1. **Productivity and cost efficiency**: automating data extraction, screening, and synthesis reduces manual analyst hours, lowers research “time-to-insight,” and scales coverage without scaling headcount (McKinsey, 2025a).
2. **Alpha generation**: by quantifying innovation signals earlier and connecting them directly to valuation inputs (growth, margins, competitive advantage duration), Jarvis increases the probability of identifying mispricing and improving entry/exit timing.

This white paper outlines (1) the market challenge and why it is particularly relevant for ODDO BHF as an active manager, (2) the investment thesis for Jarvis, (3) the technical architecture of the current prototype, and (4) a realistic roadmap for deployment, governance, and scaling.

2. The Challenge

2.1 The Background

1) The active management business model is being structurally compressed. Across Europe, assets under management (AUM) remain large and strategically important for capital allocation. According to EFAMA (2024), European asset managers are on course to manage approximately €33 trillion in assets, underscoring the sector's systemic importance. However, scale has not translated into pricing power.

Over the last decade, a combination of passive fund growth, increased transparency, and distribution pressure has driven sustained fee compression across active strategies. Broadridge (2024) documents a persistent decline in average fund fees across both US and European markets, while operating costs related to regulation, data, and technology have continued to rise.

2) The information landscape has fundamentally changed. The investment edge has shifted from "who has access to data" to "who can interpret complex data faster and more accurately." The current investment workflows still rely on comparatively simple metrics based on conventional data, such as financial reporting and broker research.

Meanwhile, numerous forward-looking signals, particularly for innovation-driven sectors, are embedded in unstructured documents, including patent filings, scientific publications, regulatory submissions, and technical standards.

McKinsey (2025b) argues that AI's primary impact in asset management will come from transforming how such unstructured data is processed and linked to decision-making, rather than from standalone predictive models.

2.2 The Problem

Problem statement: Portfolio managers lack the data access, technical capability, and scalable tooling required to systematically transform real-time, unstructured, and predictive data into valuation-relevant insights at scale.

While large volumes of potentially valuable information exist, ranging from patents and regulatory filings to news flows, sentiment, and real-world activity indicators, most asset management organizations cannot effectively utilize such data.

In practice, this manifests in four bottlenecks:

1) **Data access**: Asset managers typically lack direct, ready-to-use access to many forms of uncommon or unstructured data. Even when data sources exist, they are fragmented across public repositories, specialized providers, or internal silos, and are rarely available in a format that can be directly consumed by investment teams.
2) **Data preparation**: Raw alternative data is noisy, heterogeneous, and poorly standardized. Most investment organizations lack the specialized data engineering skills and dedicated personnel required to clean, normalize, and maintain these datasets at scale.
3) **LLM set-up**: Few asset managers have production-ready LLM infrastructures that can reliably interpret text-heavy sources, extract economically meaningful signals, and do so in a transparent, repeatable manner. Manual interpretation is infeasible given the volume and velocity of such data.
4) **Valuation integration:** Even when insights are identified, incorporating them systematically into fundamental valuation models is labor-intensive. Translating qualitative signals into adjustments to growth assumptions, margins, or risk scenarios across hundreds of securities cannot be done manually at scale within existing research workflows.

Together, these bottlenecks create a persistent and structural inefficiency: information may be available in the market, but it is not equally accessible, interpretable, or actionable across institutions.

2.3 Relevance to ODDO HBF

ODDO BHF's strategic positioning makes these challenges relevant. Oddo BHF Asset Management and Private Wealth Management both struggle to maintain outperformance with their discretionary stock picking USP. In a market where fee pressure and passive competition intensify, the highest-leverage investment is capability building that increases the probability of sustained outperformance.

Jarvis is designed for this context: an internal AI co-manager that enhances existing investment processes, respects human oversight, and aligns with governance standards expected of a European asset manager.

3. Our Solution

3.1 Why It Works

Jarvis is built on a simple but powerful premise:

Markets do not immediately price all available information—especially when that information is complex, unstructured, and forward-looking.

3.1.1 Patents are a particularly high-signal dataset

Patents have three characteristics that make them an ideal starting point for an AI-driven investment workflow:

1) **Forward-looking and predictive**: Patents are filed well before products reach market, providing early signals of future revenue potential, competitive positioning, and R&D direction.
2) **Rich text format**: Patent documents contain structured language (claims, prior art, classifications), making them particularly suitable for NLP and large language model (LLM) analysis.
3) **Quantifiable quality signals**: A substantial body of academic literature has demonstrated that citation-weighted patent measures are correlated with firm market value, productivity, and intangible capital (Hall, Jaffe, & Trajtenberg, 2004).

These properties make patents an analytically tractable proof point for how AI can transform unstructured data into valuation-relevant signals.

3.1.2 The broader universe of uncommon data

Many economically meaningful signals are embedded in uncommon, fragmented, and non-financial datasets that are not systematically integrated into investment workflows today, including:

1) Formal enterprise reporting, including annual and quarterly reports, statements, press releases, and interviews.
2) Social and stakeholder sentiment, such as discourse among customers, employees, suppliers, and communities.
3) Regulatory and administrative records, including land registries, licensing databases, and permitting activity.
4) Physical economy indicators include transportation flows, satellite or mobility proxies, and logistics congestion.

5) Digital exhaust, such as web traffic, advertising intensity, and platform activity metrics.
6) News and narrative data, including real-time news flows, regional and local media, and regulatory announcements.

Individually, these datasets are noisy. Collectively, when interpreted through semantic models and linked to valuation logic, they can reveal early shifts in demand, supply constraints, operational stress, or strategic momentum.

3.1.3 Why now: AI as an economic enabler

Historically, the barrier to using patents in investment decisions was not access but analysis cost. Modern NLP and LLM techniques reduce this cost by orders of magnitude by enabling:

- semantic understanding of technical documents
- scalable feature extraction from text
- and human-readable summarization with traceability

Industry research increasingly frames AI as a productivity lever in asset management rather than a black-box forecasting tool. McKinsey (2025a) estimates that AI-driven process transformation could materially improve cost-income ratios across financial services, reinforcing the strategic logic of investing in internal AI capabilities.

3.2 How It Works

In essence, Jarvis enhances traditional valuation models by systematically enriching them with information extracted from patents and other unstructured data.

To achieve this, the system combines patent data, market data, and financial reporting data and processes them through an AI-based interpretation layer that translates raw information into a new company valuation used in investment decision-making.

At a high level, Jarvis consists of five core building blocks:

1. **Data ingestion**: Collecting patent data, market data, and financial reporting data.
2. **Semantic extraction**: Interpreting unstructured text using large language models.
3. **Signal generation**: Converting interpretations into structured, comparable indicators.
4. **Valuation integration**: Mapping signals into multiples and DCF assumptions.

5. **Decision support**: Embedding outputs into PM workflows with human oversight.

While all five blocks are required for an end-to-end system, Jarvis' differentiation lies primarily in semantic extraction, signal generation, and valuation integration, which are described in more detail below.

1) Semantic Extraction: From Text to Economic Meaning
Unstructured sources such as patents contain economically meaningful information, but are not directly usable in investment workflows. Jarvis applies large language models (LLMs) and natural language processing techniques to interpret patent claims, abstracts, and metadata, and other unstructured data at scale.

This layer focuses on:

- Parsing and normalizing patent documents into consistent analytical units
- Classifying innovation activity into technology domains linked to business value drivers
- Distinguishing incremental developments from more substantive technological shifts.

The objective is not prediction, but interpretation: translating technical language into structured descriptions of innovation activity that can be compared across companies and over time.

2) Signal Generation: Creating Comparable Indicators
Semantic interpretations are then transformed into quantitative, comparable signals that summarize a firm's innovation profile. Examples include:

- Innovation intensity (e.g., patent activity adjusted for firm size),
- Innovation quality (e.g., citation-weighted measures),
- Technology momentum (acceleration or deceleration within specific domains),
- Competitive positioning relative to peers.

These indicators are directional and explanatory, not standalone trading signals. Their purpose is to provide a structured bridge between unstructured information and traditional financial analysis.

3) Valuation Integration: Connecting AI Signals to Investment Models
With Jarvis, AI-derived insights are not presented in isolation. They are explicitly integrated into valuation frameworks in order to compare a stock's value before and after incorporating unstructured predictive data.

Multiples-based valuation (primary use case): Innovation signals help contextualize relative valuation by informing whether a company's growth outlook, margin sustainability, or strategic positioning justifies a premium or discount compared to its peers.

Discounted cash flow analysis (secondary): Signals inform assumptions, such as revenue growth trajectories, duration of competitive advantage, and long-term reinvestment needs, supporting structured scenario analysis rather than relying on mechanical model outputs.

4) Decision Support and Human Oversight
Jarvis surfaces its outputs through a decision-support interface that highlights:

- Intrinsic value versus market price
- Key innovation-driven drivers behind valuation differences
- Contextual timing considerations using market and macro indicators.

3.3 Competitive Advantages

Jarvis is a system that transforms complex, unstructured textual data into trading signals by gauging their impact on valuations for the first time.

3.3.1 No Existing Solutions Have Our Edge

Most existing solutions in asset management fall into one of the following categories:

1) Traditional financial data platforms provide structured market and accounting data.
2) Alternative data providers offer access to unconventional datasets, typically leaving interpretation, validation, and integration to the user.
3) AI-driven analytics tools, which summarize text or generate insights but operate largely as standalone dashboards, detached from valuation logic.
4) Quantitative or factor-based systems, which optimize portfolios based on historical relationships rather than forward-looking, text-based signals.

3.3.2 Jarvis' Core Competitive Advantage

Jarvis' primary competitive advantage is that it closes the loop between unstructured information and valuation. For the first time, the system is designed to:

1) Extract and clean complex textual data at scale.
2) Contextualize data by linking it to global trends.
3) Extract economically meaningful signals from that data.
4) Translate those signals directly into valuation-relevant variables.
5) Embed the results into investment models for direct use.

6) The LLM's calculations and assumptions are numeric and thus fully tractable to PMs.

3.3.3 Why This Advantage Is Difficult to Replicate

1) Technical complexity: Interpreting unstructured text reliably, generating consistent signals, and mapping them into valuation frameworks requires a combination of NLP expertise, financial modeling knowledge, and system architecture that few teams possess simultaneously.

2) Workflow integration: Jarvis is embedded into valuation and decision workflows rather than layered on top as an external tool. This requires a deep understanding of how portfolio managers actually reason about investments.

3) Institutional learning & switching cost: Over time, feedback loops align the system with firm-specific investment styles and judgment, creating a compounding, non-transferable advantage and increasing switching cost.

3.3.4 Competition as It Will Exist in the Future

As AI adoption accelerates, more tools will emerge that analyze text, summarize documents, or surface alternative data. However, text interpretation alone is not the competitive frontier. The durable competitive advantage lies in:

turning interpretation into economically actionable valuation inputs.

4. Value Creation

Jarvis creates value by delivering new, forward-looking information that is not yet reflected in market prices and by making it directly usable for valuation and investment decision-making.

Jarvis does not aim to replace portfolio managers, automate trading, or optimize portfolios mechanically. If this information advantage is real and persistent, outperformance and AUM growth follow as consequences.

Based on ODDO BHF's current AUM profile, approximately 40% of assets are in strategies where research intensity and discretionary decision-making are most critical (fundamental equity, quantitative equity, multi-asset strategies).

Scenario-based modeling suggests that even a 1–2% net inflow effect on this addressable AUM base over time, driven by improved retention and selective inflows, would represent a meaningful commercial impact.

4.1 Short-Term Value (MVP): New Information, Made Actionable

The current MVP of Jarvis focuses on a clearly defined and concrete objective: extracting valuation-relevant signals from patent data and integrating them directly into investment analysis.

1. **Stock selection and idea generation**: Uncommon data signals highlight companies where innovation activity, technological direction, or strategic positioning is changing before this is visible in financials or consensus expectations.
2. **Valuation assessment**: The primary value of these signals lies in valuation. Jarvis funnels uncommon data interpretations directly into multiples and, secondarily, DCF analysis.

Backtesting of these signals is ongoing and not yet complete. The MVP is explicitly designed to validate signal quality, interpretability, and valuation relevance before drawing definitive conclusions on performance impact.

Taken together, our internal scenario analysis suggests that AI-enabled decision support can support a **1.5–2.5% annual excess return range over a full market cycle**, without changing investment philosophy, portfolio mandate, or governance structure. For more details, please refer to the *Appendix*.

4.2 Long-Term Value (Jarvis's Vision): AUM Stability and Growth

In asset management, performance improvements do not translate instantly into inflows. The transmission mechanism runs through performance persistence, client confidence, and retention.

Over the long term, the primary value of Jarvis lies in strengthening the firm's structural position, rather than in any single performance metric. Key long-term effects include:

1) **Continuous learning**: As Jarvis learns from internal outcomes, feedback loops, and portfolio results, decision quality improves over time, with the strongest gains typically occurring in early years and compounding thereafter.
2) **Effective use of uncommon and alternative data**: Jarvis enables scalable integration of non-traditional datasets—beyond patents alone—allowing the firm to extract insight from sources competitors struggle to operationalize.
3) **Alpha protection**: By benchmarking new decisions against historical precedents and highlighting analogous past outcomes, Jarvis helps avoid large, avoidable drawdowns and repeated mistakes.

Within ODDO BHF's current portfolio, this primarily includes fundamental equity strategies (approximately 11% of AUM), quantitative equity strategies (approximately 9%), and multi-asset strategies (approximately 20%). Taken together, these segments represent roughly 40% of total assets under management, where AI-enabled decision support can have a meaningful impact.

To illustrate the order of magnitude, an indicative scenario assumes that AI-enabled improvements affect the addressable AUM base of approximately €25.6 billion (40% of the total AUM of €64 billion). Under this scenario, even a modest 4% uplift in assets over time—driven by improved retention and incremental inflows—would correspond to approximately **€1.0 billion in additional AUM**. This illustration is not a forecast, but a representation of how relatively small, sustained improvements in investment outcomes can compound into material commercial effects over time.

5. Technical Explanation

5.1 The Current Prototype

What it does today: The current Jarvis prototype is a fully functional MVP designed to validate the end-to-end analytical workflow, user experience, and performance characteristics of an AI-enabled co-manager.

Speed (current measurement): For a 5-ticker portfolio, an end-to-end run completes in approximately 20-30 seconds. This performance is achieved through parallel processing, utilizing a **Hybrid Multi-Source Engine** to maximize throughput. The system intelligently rotates between Yahoo Finance, FinViz, and AlphaVantage, employing randomized throttling (5-20 seconds) and native session management to mitigate rate limits and ensure successful data ingestion.

What a user gets at the end of a run: A clean, structured dashboard based on the entire portfolio—taking insights from financial tickers (Yahoo Finance) and patent data (PatentsView for US patents). The user sees:

1) Patent innovation analysis: AI-generated assessments of innovation quality (1-10 scale), commercial readiness (1-10), and litigation risk (1-10) with R&D trend identification (e.g., "Pivoting from consumer electronics to spatial computing and health tech ecosystem (2021-2024)).
2) Investment Strategy Analysis: Portfolio analysis calculates cumulative returns, Sharpe ratios, and maximum drawdown, alongside a Fama-French three-factor regression (based on Kenneth R. French) to decompose returns into market exposure (beta), the size factor (SMB), and the value factor (HML). This directly supports the "Risk Awareness" value driver (Section 3.1) by stress-testing new ideas against historical factors.
3) Unified Recommendations: A final AI synthesis that combines financial metrics, patent analysis, and analysis results into unified investment recommendations, reducing the "Synthesis Latency" that typically delays PM decision-making.
4) AI Deep Dive: Displays in-depth analysis on the portfolio as a whole with both bull and bear cases, strategic trajectories by holding, and individual company fundamentals.

5.2 Technical Setup

5.2.1 High-level architecture

The setup separates orchestration, computation, and storage. A workflow orchestrator coordinates and automates each step, and a managed PostgreSQL database triggers automatically to coordinate workflow execution by fanning out a portfolio of individual ticker requests. This separation keeps the UI fast and enforces the "governance-ready design" discussed in Section 2.3.

Financial data ingestion (FDI) and patent data ingestion (PDI) run in parallel for all tickers. Once both FDI and PDI are complete for the entire portfolio, a database trigger automatically fires the backtesting workflow (including FFM analysis). Finally, when analysis routines are complete, another trigger initiates the AI synthesis workflow to generate the portfolio-level bull and bear cases.

Computation happens inside Docker containers hosted on AWS servers to keep environment dependencies consistent while keeping an eye out for scalability. Finally, the frontend reads only the final joined results needed for display.
This database-driven orchestration ensures zero human intervention and sub-minute latency between workflow stages. database, stores inputs/outputs, and run status.

Fig. 1 scanning the QR below illustrates the complete end-to-end flow from portfolio creation through final AI synthesis.



5.2.2 Core infrastructure components

1) N8N (Workflow Orchestration): Self-hosted on unisaarland.app.n8n.cloud, running 4 primary workflows: Dispatcher, Financial Data Ingestion (FDI), Patent Data Ingestion (PDI), and Backtest + FFM Analysis, and final AI Synthesis.
2) Supabase (Database): Managed PostgreSQL instance with pg_net extension enabled for webhook triggering, providing ACID transactions and real-time subscriptions.
3) Jarvis Market Data Service: FastAPI application (Python 3.9+) running in Docker on AWS EC2 instance. It features a ‘Multi-Layer' architecture that

orchestrates data fetching across primary sources (FinViz, Yahoo) and a Fail-Safe Backup (AlphaVantage) to guarantee data integrity even during provider outages.

4) Backtest Engine: A separate FastAPI service also running in Docker on an AWS EC2 instance, handling Fama-French factor analysis and strategy backtesting using vectorized NumPy/Pandas computations.
5) Frontend: Lovable-built React application with real-time portfolio status polling and interactive dashboard visualization.

All services communicate via REST APIs and database webhooks, with no direct coupling between components—enabling independent scaling and updates.

How the system stays organized: Each portfolio run is tracked using UUIDs across four interconnected database tables:

- portfolios: Stores portfolio-level metadata and aggregated status (fdi_status, pdi_status, backtesting_status, ai_synthesis_status, portfolio_status).
- track_requests: One row per ticker, tracking individual ticker progress (financial_analysis_status, patent_analysis_status, backtest_status, ticker_status).
- ticker_data: Stores raw OHLCV price data (via native Yahoo finance) and company fundamentals enriched by a hybrid scraper that fuses data from FinViz, Yahoo, and AlphaVantage to eliminate null values.
- ffm_results & backtest_results: Stores quantitative analysis outputs (alpha, beta, Sharpe ratio, etc.) Instead of relying on human judgment, completion is recorded atomically in the database using BEFORE and UPDATE triggers. When all tickers reach 'complete' status, the portfolio automatically transitions to 'finished,' signaling the frontend to display results.
- Patent_data: Stores both the raw patent count and AI-structured analysis as a JSON payload (summary_payload). The AI transforms unstructured patent records into investment-relevant metrics—see "AI-powered patent structuring" below.

This approach prevents "race conditions" and ensures that the "Decision Consistency" promised in Section 3.3 is mechanically enforced by the system.

Below is the Entity Relationship Diagram that can be viewed by scanning the QR code (Fig2):



5.2.3 AI-powered patent structuring (PDI workflow)

PatentsView returns unstructured data, including patent titles, abstracts, filing dates, and assignee names. The PDI workflow sends this raw data to OpenAI GPT-4 Mini with a structured prompt asking the AI to analyze the patent portfolio across five investment-relevant dimensions:

1) **Innovation Quality (1-10)**: Are these breakthrough inventions or incremental improvements? The AI examines claim scope, novelty indicators, and technical depth.
2) **Commercial Readiness (1-10)**: How close are these patents to generating revenue? The AI considers the specificity of the claims, the practical applications described, and the potential for industry adoption.
3) **Litigation Risk (1-10)**: The AI assesses overlap with crowded patent spaces, potential troll targets, or defensive weaknesses based on claim overlap analysis.
4) **Moat Width**: The AI categorizes the IP portfolio as Narrow (easy to design around), Wide (strong protection), or Fortress (industry-defining IP) by analyzing claim breadth and citation patterns.
5) **Strategic Trend**: The AI analyzes patent chronology to identify the company's R&D trajectory. Examples of AI-generated trends:
    a) "Pivoting from consumer hardware to enterprise AI infrastructure (2020-2024)."
    b) "Doubling down on battery tech while abandoning autonomous systems (2018-2023)."
    c) "Expanding from social media to the VR/AR metaverse (2021-2024)."

The AI's structured output is stored in patent_data.summary_payload as JSON, making it immediately queryable and displayable in the dashboard. This transformation from "147 unstructured patent records" to "Innovation Score: 8.5, Strategic Trend: AI infrastructure pivot" is what makes patent data actionable for investment decisions without requiring users to read hundreds of abstracts.

Why AI here? Patent analysis is a natural language understanding task. Identifying trends, assessing novelty, and predicting commercial impact require contextual reasoning that traditional keyword searches cannot provide. By converting these records into "Strategic Trend: AI Pivot," the system validates the "Entry Wedge" thesis (Section 2.1)—proving that complex, unstructured text can be turned into quantifiable valuation signals.

Below is a Workflow-sequence architecture represented with QR code (Fig3).



Fig. 3 illustrates the complete end-to-end flow from portfolio creation through final AI synthesis. Key architectural highlights include: - Automatic fan-out from 1 portfolio to N ticker requests (Phase 1) - Parallel execution of FDI and PDI workflows (Phase 2) - Database-triggered synchronization gate ensuring backtest only runs after both FDI and PDI are complete (Phase 3) - Final AI synthesis generating portfolio-level investment thesis (Phase 4) This event-driven architecture eliminates manual workflow triggering and achieves sub-minute latency between phases.

Key design choices (kept deliberately simple for MVP):

- Parallel processing: Per-ticker jobs and backtesting calculations run simultaneously to reduce wall-clock time from minutes to seconds.
- Structured AI outputs: Patent analysis and strategy insights are stored as JSON with a defined schema (innovation_score, strategic_trend, alpha_generation, etc.), so the UI can display metrics without reinterpreting text.
- Single source of truth: The database stores both statuses and results to avoid mismatches between services.
- Atomic completion checks: Portfolio-level status updates occur within single database transactions to prevent race conditions, further protected by processes that run at the end of each n8n workflow.

Error resilience strategies (MVP): While comprehensive failure handling is planned for Phase 1, the current MVP includes several defensive mechanisms:

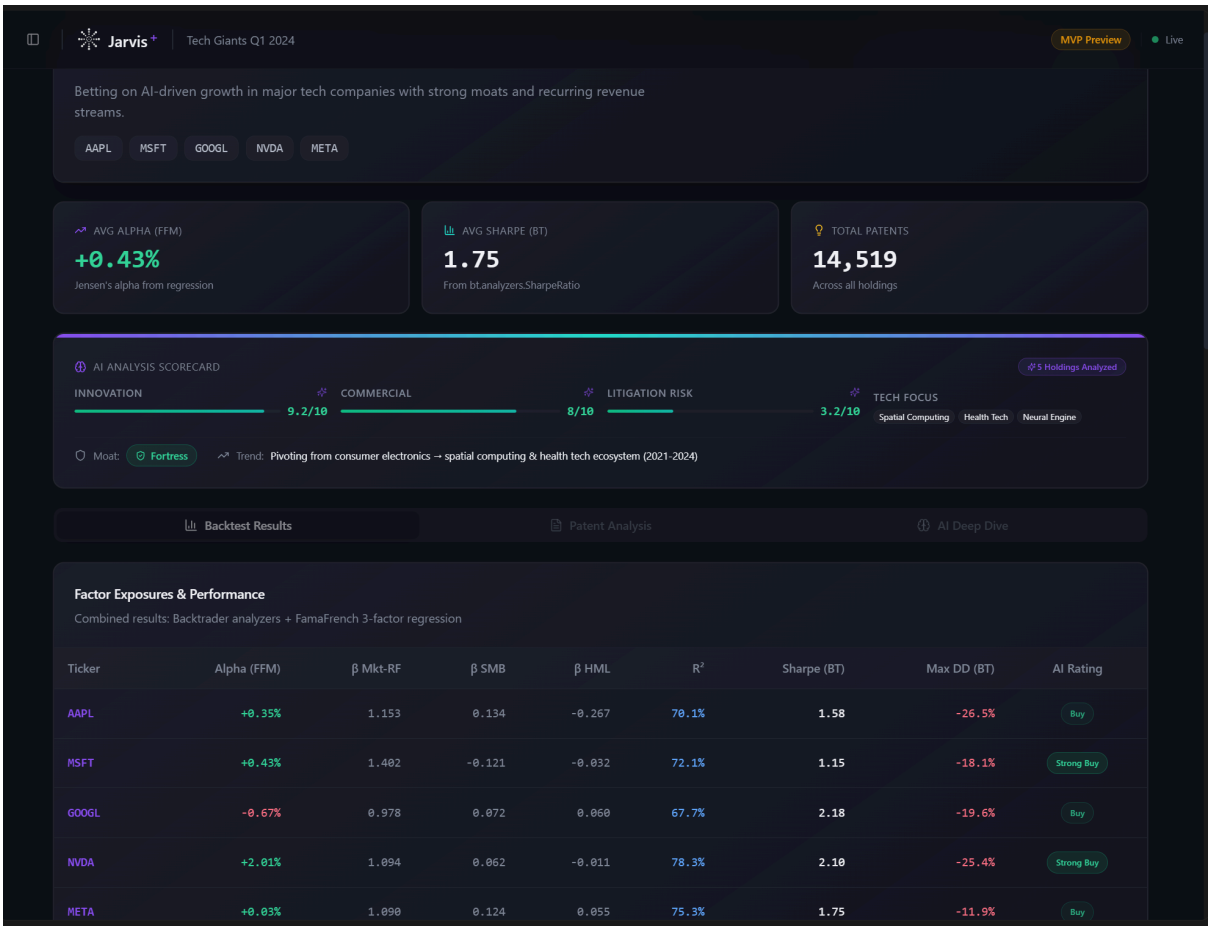
- Retry logic: Yahoo Finance requests automatically retry up to 3 times with exponential backoff (3s, 6s, 12s) before marking a ticker as failed.

- User-Agent rotation: Randomized browser headers reduce the likelihood of being flagged as a bot by data providers.
- Graceful degradation: If patent data fails for a ticker, the analysis continues with financial data alone, flagging the missing component in the error_log field.
- Status tracking: Every ticker maintains independent status fields, so one failure doesn't cascade to other tickers in the portfolio.
- Timeout protection: All HTTP requests include 20-second timeouts to prevent indefinite hangs.

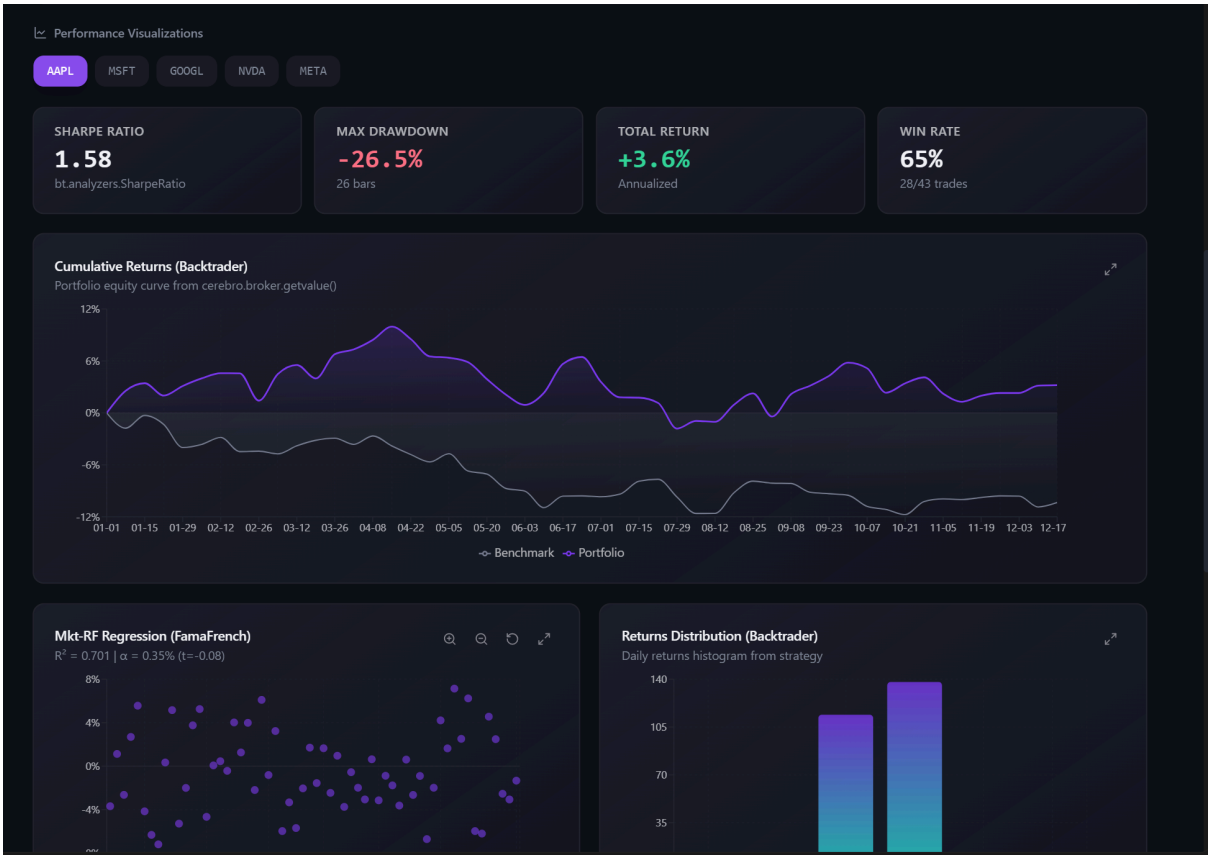
Human Oversight: AI is used to structure and summarize information and, more importantly, assist humans at play. Patent analysis outputs include clear metrics (scores 1-10, defined risk categories) and direct quotes from the AI's reasoning. Every result can be reviewed, compared across portfolios, and rerun with different parameters. This keeps the system credible for risk-averse reviewers because it is transparent about what was computed, what was fetched, and what was inferred.

Below are some sample screenshots with dummy data to test the look and feel (final view subject to changes):

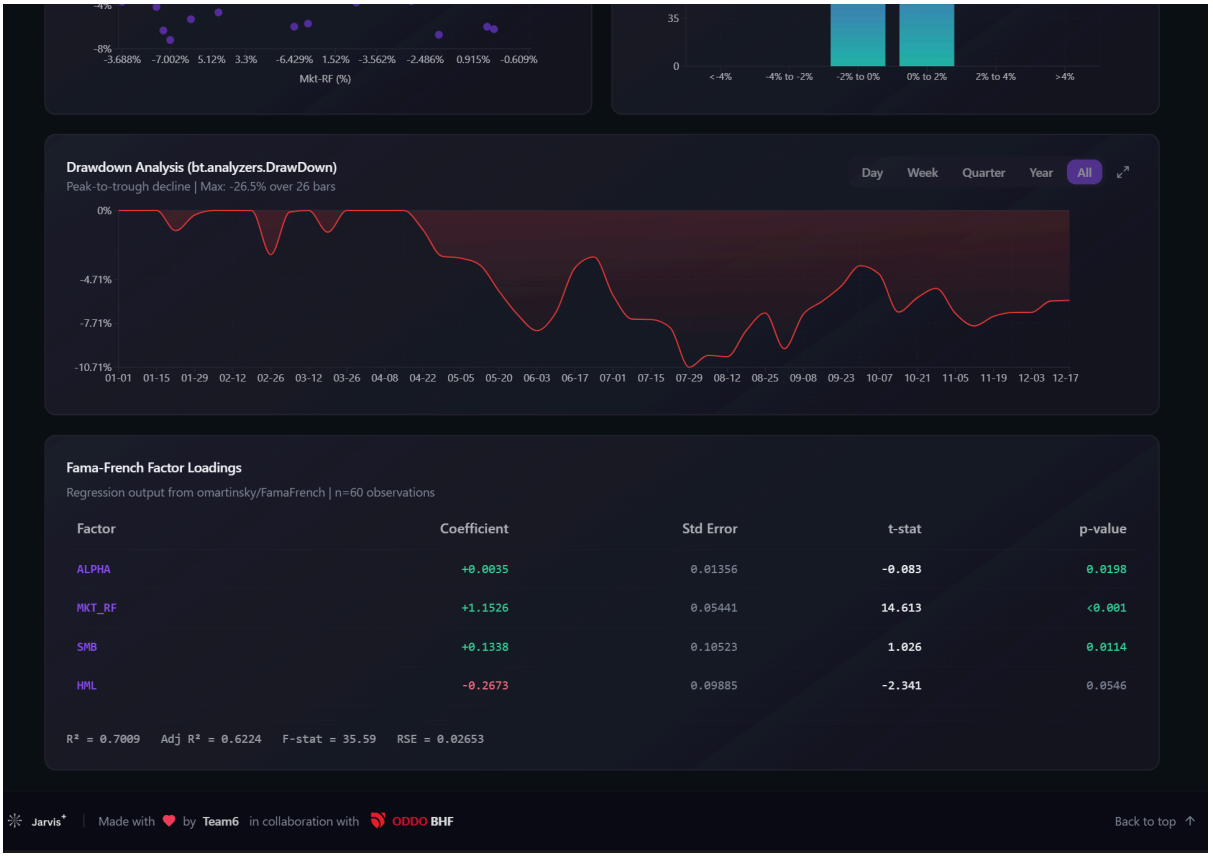
Screenshot1:



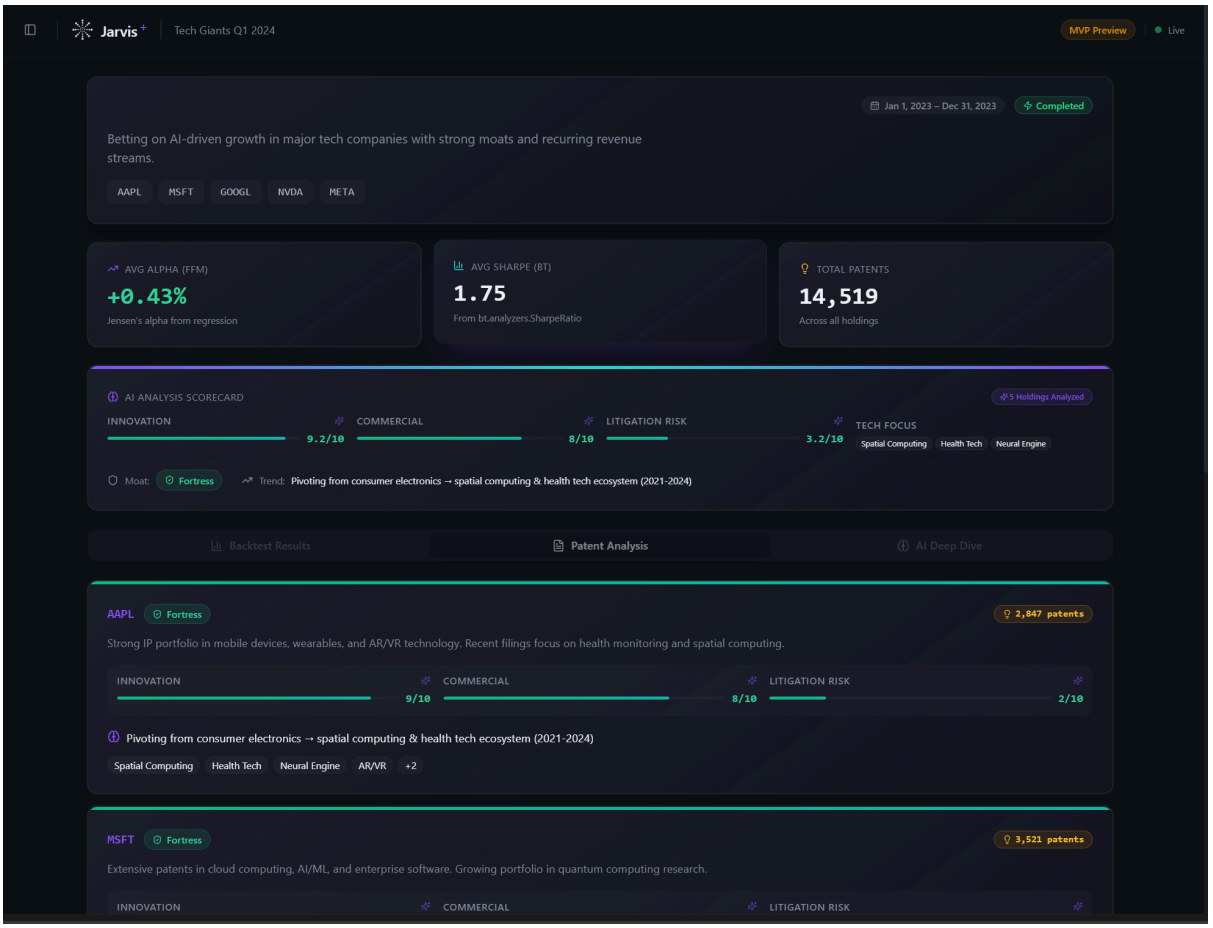
Screenshot 2:



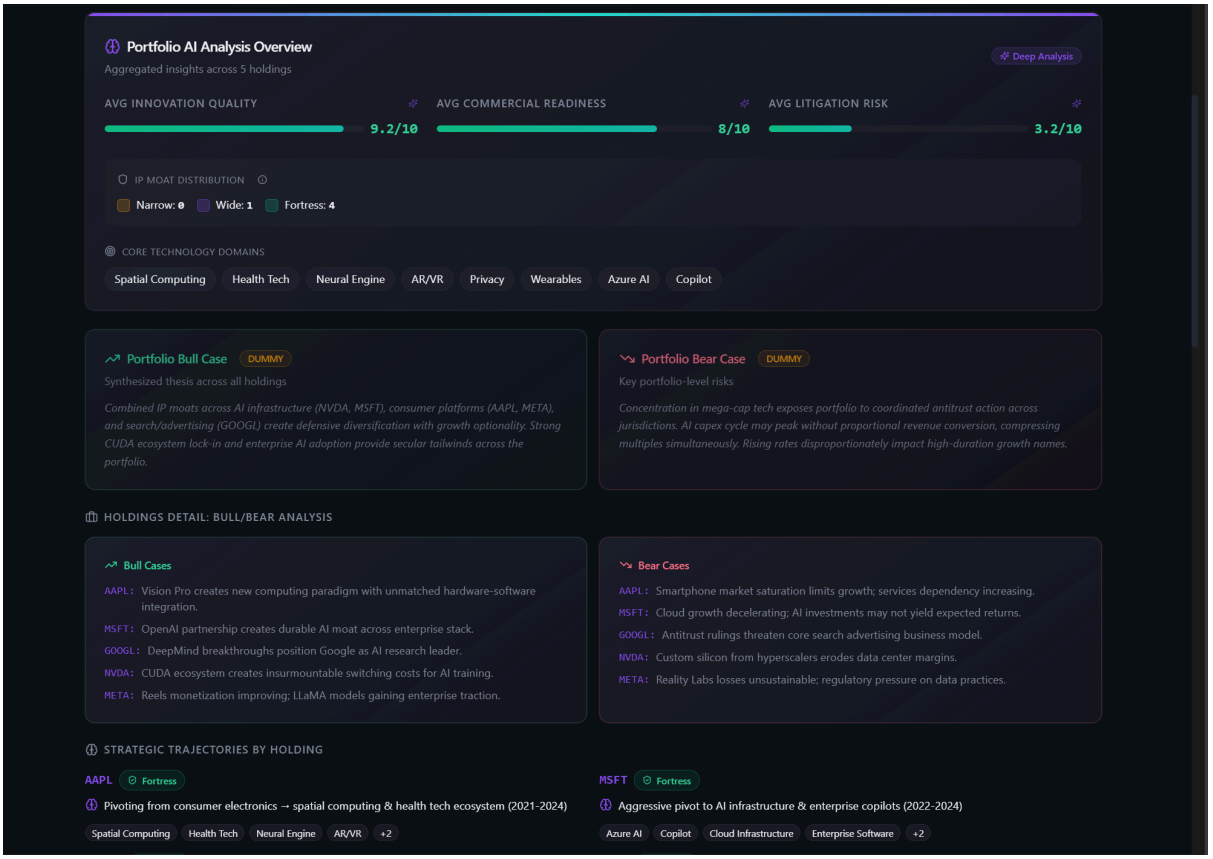
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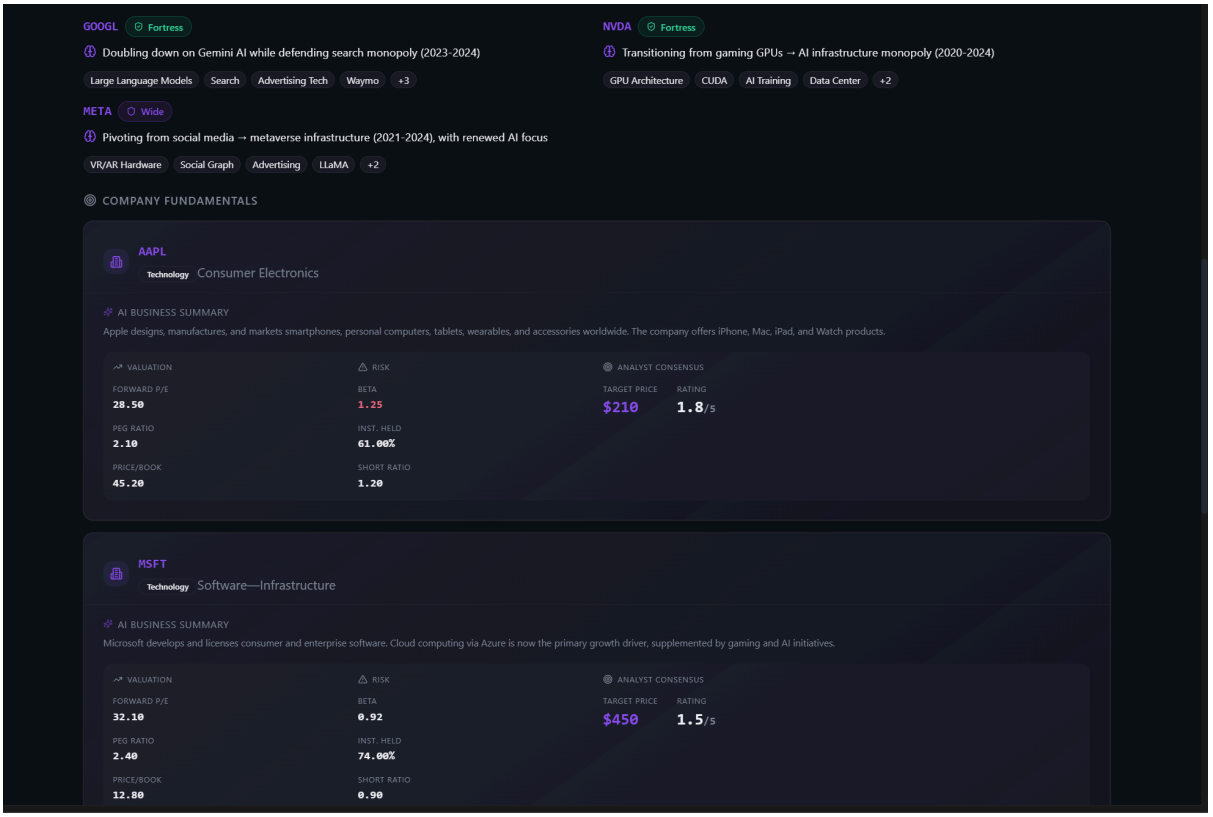

-8%
-3.688%
-7.002%
5.12%
3.3%
-6.429%
1.52%
-3.562%
-2.486%
0.915%
-0.609%
Mkt-RF (%)
35
0
<-4%
-4% to -2%
-2% to 0%
0% to 2%
2% to 4%
>4%
Drawdown Analysis (bt.analyzers.DrawDown)
Peak-to-trough decline | Max: -26.5% over 26 bars
Day
Week
Quarter
Year
All
0%
-4.71%
-7.71%
-10.71%
01-01
01-15
01-29
02-12
02-26
03-12
03-26
04-08
04-22
05-05
05-20
06-03
06-17
07-01
07-15
07-29
08-12
08-25
09-08
09-23
10-07
10-21
11-05
11-19
12-03
12-17
Fama-French Factor Loadings
Regression output from omartinsky/FamaFrench | n=60 observations
Factor
Coefficient
Std Error
t-stat
p-value
ALPHA
+0.0035
0.01356
-0.083
0.0198
MKT_RF
+1.1526
0.05441
14.613
<0.001
SMB
+0.1338
0.10523
1.026
0.0114
HML
-0.2673
0.09885
-2.341
0.0546
R² = 0.7009
Adj R² = 0.6224
F-stat = 35.59
RSE = 0.02653
Jarvis+
Made with
by Team6
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Screenshot 4:



Screenshot 5:


Portfolio AI Analysis Overview
Aggregated insights across 5 holdings
Deep Analysis
AVG INNOVATION QUALITY
9.2/10
AVG COMMERCIAL READINESS
8/10
AVG LITIGATION RISK
3.2/10
IP MOAT DISTRIBUTION
Narrow: 0
Wide: 1
Fortress: 4
CORE TECHNOLOGY DOMAINS
Spatial Computing
Health Tech
Neural Engine
AR/VR
Privacy
Wearables
Azure AI
Copilot
Portfolio Bull Case
DUMMY
Synthesized thesis across all holdings
Combined IP moats across AI infrastructure (NVDA, MSFT), consumer platforms (AAPL, META), and search/advertising (GOOGL) create defensive diversification with growth optionality. Strong CUDA ecosystem lock-in and enterprise AI adoption provide secular tailwinds across the portfolio.
Portfolio Bear Case
DUMMY
Key portfolio-level risks
Concentration in mega-cap tech exposes portfolio to coordinated antitrust action across jurisdictions. AI capex cycle may peak without proportional revenue conversion, compressing multiples simultaneously. Rising rates disproportionately impact high-duration growth names.
HOLDINGS DETAIL: BULL/BEAR ANALYSIS
Bull Cases
AAPL: Vision Pro creates new computing paradigm with unmatched hardware-software integration.
MSFT: OpenAI partnership creates durable AI moat across enterprise stack.
GOOGL: DeepMind breakthroughs position Google as AI research leader.
NVDA: CUDA ecosystem creates insurmountable switching costs for AI training.
META: Reels monetization improving; LLaMA models gaining enterprise traction.
Bear Cases
AAPL: Smartphone market saturation limits growth; services dependency increasing.
MSFT: Cloud growth decelerating; AI investments may not yield expected returns.
GOOGL: Antitrust rulings threaten core search advertising business model.
NVDA: Custom silicon from hyperscalers erodes data center margins.
META: Reality Labs losses unsustainable; regulatory pressure on data practices.
STRATEGIC TRAJECTORIES BY HOLDING
AAPL
Fortress
Pivoting from consumer electronics → spatial computing & health tech ecosystem (2021-2024)
Spatial Computing
Health Tech
Neural Engine
AR/VR
+2
MSFT
Fortress
Aggressive pivot to AI infrastructure & enterprise copilots (2022-2024)
Azure AI
Copilot
Cloud Infrastructure
Enterprise Software
+2


GOOGL
Fortress
Doubling down on Gemini AI while defending search monopoly (2023-2024)
Large Language Models
Search
Advertising Tech
Waymo
+3
NVDA
Fortress
Transitioning from gaming GPUs → AI infrastructure monopoly (2020-2024)
GPU Architecture
CUDA
AI Training
Data Center
+2
META
Wide
Pivoting from social media → metaverse infrastructure (2021-2024), with renewed AI focus
VR/AR Hardware
Social Graph
Advertising
LLaMA
+2
COMPANY FUNDAMENTALS
AAPL
Technology
Consumer Electronics
AI BUSINESS SUMMARY
Apple designs, manufactures, and markets smartphones, personal computers, tablets, wearables, and accessories worldwide. The company offers iPhone, Mac, iPad, and Watch products.
VALUATION
FORWARD P/E
28.50
PEG RATIO
2.10
PRICE/BOOK
45.20
RISK
BETA
1.25
INST. HELD
61.00%
SHORT RATIO
1.20
ANALYST CONSENSUS
TARGET PRICE
$210
RATING
1.8/5
MSFT
Technology
Software—Infrastructure
AI BUSINESS SUMMARY
Microsoft develops and licenses consumer and enterprise software. Cloud computing via Azure is now the primary growth driver, supplemented by gaming and AI initiatives.
VALUATION
FORWARD P/E
32.10
PEG RATIO
2.40
PRICE/BOOK
12.80
RISK
BETA
0.92
INST. HELD
74.00%
SHORT RATIO
0.90
ANALYST CONSENSUS
TARGET PRICE
$450
RATING
1.5/5

Why a Hybrid Open-Source Stack for MVP, and what's next? A Hybrid Multi-Source Engine (combining Yahoo Finance, FinViz, and AlphaVantage) was chosen deliberately for rapid MVP development. This approach validated the system's "Nuclear Option" resilience logic without incurring licensing costs or contractual delays. However, open-source scraping is not a production-grade solution for institutional asset management.

Once pilot funding is secured, the system is designed to seamlessly switch to ODDO-supported infrastructure or a more reliable, professionally tested subscription:

- **Bloomberg Terminal API or Refinitiv Eikon**: Providing industry-standard real-time data, 20+ years of historical depth, and contractual SLAs.
- **Quandl (Nasdaq Data Link)**: For specialized alternative datasets and economic indicators.

The architecture is data-source agnostic; switching from the current open-source hybrid layer to a Bloomberg feed requires only updating the Jarvis Market Data Service’s API connectors, not rewriting the workflow or database schema. The MVP proves the capability; Phase 1 delivers the infrastructure.

6. Roadmap: From MVP to a commercial product

The roadmap evolves Jarvis from a technical proof-of-concept into a commercially viable product for institutional deployment.

6.1 MVP 0: Current Foundation (Validation)

Objective: Prove the core outcome first. A fast, repeatable portfolio run that produces consistent outputs and a clear completion status. The system already prioritizes speed through parallel processing.

The system currently prioritizes speed through parallel processing and "Data Continuity" via the new hybrid multi-source architecture.

1) **AI-Structured Patent Signals**: We ingest raw patent records from PatentsView and use an AI layer to convert them into structured, investment-relevant signals—specifically Innovation Quality, Commercial Readiness, and Litigation Risk.
2) **Actionable Outputs**: The system stores outputs such as patent count, filing/citation trends, and a moat-strength tier (Narrow, Wide, or Fortress) using a VRIO-inspired defensibility lens.
3) **Thesis Generation**: Finally, within the selected time window, the system generates a patent-driven bull case and bear case for the company.

6.2 MVP 1: Valuation Logic Integration

Objective: Bridge the gap between qualitative signals and quantitative value.

Integration Logic: Rather than stopping at a "score," this phase links AI outputs directly to fundamental valuation drivers (DCF and Multiples). For example, a "High Innovation Quality" score maps to a premium on Revenue Growth assumptions, while a "Fortress Moat" rating justifies a higher Terminal Value multiple.

Note: The schematic flow for this integration is detailed in Section 3.2.

6.3 MVP 2: Systematic Backtesting & Signal Validation

Objective: Validate that the AI signals actually predict market returns (Alpha).
This is a critical validation step. We do not just backtest the portfolio; we backtest the predictive power of the signal itself to ensure it adds value beyond standard factors.

The Validation Loop:

1) **Lagged Signal Generation**: To strictly prevent forward-looking bias, the AI analyzes patent data from a historical 3-year lag (e.g., analyzing 2020 patents to predict 2021-2023 returns) without access to subsequent market data.
2) **Factor Regression**: We regress the patent signal against subsequent stock returns, controlling for standard Fama-French factors (Size, Value, Momentum) to isolate pure alpha.
3) **Feedback Mechanism**: If a signal (e.g., "Litigation Risk") shows low correlation with realized returns, its weight in the recommendation engine is automatically reduced.

6.4 MVP 3: Data Expansion & Technical Maturity

Objective: Broaden the signal funnel and harden the infrastructure for scale.

Data Source Expansion:

1) **Formal Enterprise Reporting**: Ingesting annual/quarterly reports, press releases, and interviews to detect divergence between management narrative and financial reality.
2) **Social & Stakeholder Sentiment**: Monitoring discourse among customers, employees, suppliers, and communities to capture reputation risk and momentum.
3) **Regulatory & Administrative Records**: Integrating land registries, licensing databases, and permitting activity for early economic signals.
4) **Physical Economy Indicators**: Tracking transportation flows, satellite mobility proxies, and logistics congestion to identify supply chain bottlenecks.
5) **Digital Exhaust**: Analyzing web traffic, advertising intensity, and platform activity metrics to gauge real-time demand.
6) **Market Benchmarking**: Integrating indices (e.g., MSCI World) for relative performance tracking.

Operational Enhancements:

1) **AI Response Caching**: Caching validated analysis outputs to reduce API costs and guarantee consistency across portfolio runs.
2) **Automated Evaluation**: Maintaining a representative test suite to benchmark model performance and accuracy before every update.

6.5 Institutional Readiness (The Scalable Platform)

Objective: Deployment, Governance, and User Experience.

This phase combines previously planned "Institutional Readiness" and "Scalable Platform" features into a single release for ODDO BHF professionals.

1) **Proprietary Models (Privacy)**: Transitioning to internal Small Language Models (SLMs) hosted by ODDO BHF. This ensures zero data leakage and allows strict control over training data to eliminate forward bias.
2) **Multi-Market Expansion**: Extend coverage to major EU exchanges (Euronext, Deutsche Börse, LSE) and Asian markets, handling currency conversions and regional data availability.
3) **Workflow Enhancements**:
    a) Compare Portfolios: Side-by-side results to justify choices and spot outliers.
    b) Trend Detection: Surface R&D trajectories and provide forecast ranges to speed up screening.
    c) Robust Error Handling: Automatic retries for transient failures and dead-letter tracking for stuck workflows.
    d) Export/Download: Shareable PDF reports for offline review and archiving.

7. Team Composition

We are a motivated and diverse team that combines financial knowledge, technical rigor, and sharp business execution. We are in the best position to push Jarvis to the next phase with continual impact.

Violet Si: Operational & Commercial Lead

Violet leads the operational and commercial development of Jarvis, ensuring that the initiative translates from a technical concept into a deployable, governance-ready capability. She focuses on defining high-impact use cases, structuring the operating model, and aligning product development with portfolio manager workflows and organizational constraints. Her role also involves coordinating across the team and shaping a coherent strategic and commercial roadmap.

Mohamed Raslan: Financial Lead

Mohamed is responsible for the financial and economic analysis underpinning Jarvis. He evaluates value-creation levers, including alpha generation, AUM dynamics, cost efficiency, and scalability. His work ensures that the initiative's financial logic is robust, realistic, and aligned with stakeholder expectations, with a strong focus on long-term sustainability and risk-aware value creation.

Sarbajit Chatterjee: Product Architect

Sarbajit serves as the Product Architect, translating strategic objectives into a coherent and scalable product design. He defines system structure, user workflows, and feature prioritization to ensure that technical development remains closely aligned with real investment processes. His role is supporting Jarvis's evolution from a prototype to an institutionally credible solution.

Rushan Mukherjee: AI Expert

Rushan provides expertise in applied AI and machine learning, focusing on backtesting and how advanced models can be used to extract and structure insights from unstructured data in a transparent and explainable manner. He ensures that AI components support human decision-making rather than operate as black-box systems, and that model design choices remain robust, adaptable, and suitable for use in a regulated asset management environment.

Dr. Ingolf Pernice: Industry Advisor

Ingolf acts as an Industry Advisor, providing ongoing guidance grounded in real-world asset management practice. Through regular engagement with the team, he challenges assumptions, validates priorities, and ensures alignment with ODDO BHF's investment philosophy and operational realities. His input helps ensure that Jarvis remains institutionally relevant and practically adoptable rather than purely theoretical.

8. Conclusion: The Investment Thesis

Jarvis is a focused investment in a small number of structurally powerful capabilities that directly address the core challenges of active asset management. While many potential applications of AI exist, the investment value of Jarvis rests on three pillars.

Pillar 1: Uncommon, Forward-Looking Data as a Source of Differentiation
At the heart of Jarvis is the systematic use of uncommon, forward-looking data, with patents serving as the initial and most analytically tractable proof point.

The MVP deliberately focuses on this pillar, proving that such data can be reliably extracted, interpreted, and structured at scale. If this capability works in backtesting, it establishes a durable informational edge that is difficult to replicate using traditional datasets alone.

Pillar 2: Direct Integration into Valuation
The second core pillar is valuation integration. Jarvis funnels interpreted signals directly into valuation frameworks that portfolio managers already use, most notably relative multiples, to support deeper reasoning and discussion.

By embedding AI-derived insights into growth assumptions, competitive advantage duration, and scenario analysis, Jarvis ensures that interpretation is translated into economically actionable variables, not abstract indicators. MVP 1 explicitly builds on this pillar, extending and validating the valuation linkage and subjecting it to structured backtesting.

Pillar 3: Institutional Learning and Compounding Advantage
The final pillar ensures that the edge created by pillars 3 and 4 does not decay over time. Through feedback loops, outcome tracking, and decision benchmarking, Jarvis becomes progressively aligned with the institution's investment style, risk appetite, and strategic priorities.

A Focused Scope
Importantly, Jarvis does not attempt to solve every problem in asset management. The investment case stands on a simple proposition:

- If uncommon, forward-looking data can be interpreted reliably at scale, and
- If those interpretations can be embedded directly into the valuation logic used by portfolio managers,

Then a durable decision advantage exists. Everything else is supportive, not essential. For an institution such as ODDO BHF, Jarvis represents a disciplined, economically grounded response to structural industry pressures.

9. Special Thanks

At the end, we would like to say a special thank you to the entire Triathlon-SOUTHWESTX Hackathon team and to senior members from ODDO HBF for your support and insight, without which none of this would have been possible.

10. Appendix

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10.2 Detailed Breakdown of Value Creation Logic and Calculation

Alpha Generation (Illustrative Breakdown)

Jarvis supports alpha generation by improving multiple, overlapping dimensions of the investment decision process. The following breakdown illustrates how incremental improvements across these dimensions can compound at the portfolio level:

- Stock Selection (+1.2%)
  Improved idea generation through the integration of fundamental analysis, alternative data, and machine learning–based pattern recognition, resulting in higher hit rates and more consistent security selection.
- Timing and Information Speed (+0.6%)
  Faster interpretation of earnings releases, macroeconomic news, and sentiment shifts enables earlier validation or rejection of investment signals, without reliance on high-frequency trading.
- Risk Management (+0.4%)
  Enhanced downside-risk detection, improved drawdown control, and more dynamic position sizing contribute to superior risk-adjusted performance.
- Portfolio Construction (+0.3%)
  More accurate correlation estimates, reduced unintended factor exposure, and improved diversification efficiency at the portfolio level.

Taken together, these effects support an illustrative target of approximately +2.5% annual excess return, achieved without changes to investment philosophy, portfolio mandates, or governance structures. The contribution arises from enhanced decision quality rather than strategy alteration.

Assets Under Management (AUM) Growth

Total AUM Context

- Total AUM: €64 billion
- AUM is the primary driver of recurring revenue, as management fees scale directly with assets under management.

AI-Addressable AUM Scope

AI-driven decision support is most impactful in strategies where discretionary judgment, research intensity, and performance persistence matter most:

- Fundamental Equity: 11%
- Quantitative Equity: 9%
- Multi-Asset Strategies: 20%

Total AI-addressable AUM: 40% of total AUM

Rationale for AUM Impact

AI-enabled improvements in:

- risk-adjusted performance,
- consistency of outcomes,
- and decision speed and quality

translate into:

- higher client retention,

- incremental net inflows,
- and stronger long-term AUM stability.

Illustrative First-Year Scenario

- Addressable AUM: €64bn × 40% = €25.6bn
- Assumed AI-driven uplift: 4%
- Indicative incremental AUM: €1.024bn

Cost Efficiency and Productivity

Cost per Investment Analysis

- Analyst time: reduced, but reallocated to higher-value activities
- Data processing: largely automated
- Review and validation: faster, AI-assisted

This supports an illustrative ~19% structural reduction in research cost per decision.

Capacity Reallocation Effects

Beyond direct cost savings, Jarvis unlocks productivity gains by shifting analyst effort toward:

- idea generation,
- risk assessment,
- portfolio construction,
- and client-facing explanations.

This effectively increases team capacity by 15–20% without additional hiring.

Summary Impact Across Dimensions

Dimension	Illustrative AI Impact
AUM	↑ Strong Growth
Alpha	↑ Consistent Excess Return
Costs	↓ Structural Reduction
Productivity	↑ Significant
Risk	↓ Better Control
Client Value	↑ High
Firm Valuation	↑↑ Long-Term

Competitive Advantage

AI adoption strengthens competitiveness by improving portfolio quality, decision consistency, and execution discipline. Estimated impacts include:

- 10–20% improvement in decision consistency across portfolios
- higher client retention driven by performance stability

Reduction of Structural Weaknesses in Investment Planning

Jarvis helps identify and reduce recurring weaknesses by:

- highlighting gaps between strategy and execution,
- stress-testing assumptions before capital deployment,
- detecting repeated planning and execution errors.

Illustrative impact:

- 15–30% reduction in recurring decision errors

Continuous Learning Effects

As the system learns from internal outcomes, decision quality improves over time in alignment with ODDO BHF's:

- investment style,
- risk appetite,
- and strategic priorities.

Illustrative impact:

- 5–10% annual improvement in signal accuracy in early years, with compounding effects thereafter.

Use of Uncommon and Alternative Data

Jarvis enables systematic use of uncommon and unstructured data, including internal notes, qualitative assessments, and narrative information that traditional models cannot efficiently process.

Illustrative impact:

- 0.5–1.0% incremental alpha contribution annually

Decision Benchmarking and Alpha Protection

By benchmarking new investment cases against historical decisions—drawing on research archives, PM rationales, and risk committee notes—Jarvis helps avoid repeated mistakes.

Illustrative impact:

- 20–30% reduction in large, avoidable drawdowns

Reduction of Behavioral Biases

As an independent reviewer, AI reduces behavioral biases such as overconfidence and confirmation bias, strengthening fiduciary discipline.

Illustrative impact:

- 10–15% improvement in risk-adjusted returns (Sharpe ratio)

Scalable Expansion Across Functions and Markets
Jarvis supports expansion across investment management, risk, compliance, and client advisory without linear increases in headcount.

Illustrative impact:

- 20–30% scalability gain without proportional cost growth

10.3 Current Limitations & Assumptions

Happy-path flow (by design): The MVP prioritizes a smooth successful run over comprehensive failure/error handling. Today, the focus is on the core outcome. It needs to be fast and consistent with the results for valid inputs. Thus aligning with the "Phase 0" goal of validating signal quality before optimizing for "Operational Leverage."

Advanced error handling (automatic retries for API failures, dead-letter queues for stuck workflows, and proactive alerting for rate limits) will be introduced in Phase 1 after the core pipeline proves stable under real-world usage.

Known data source constraints:

- Yahoo Finance (via yfinance library in Python):
    - Historical financials limited to 4 years/quarters: Balance sheets, cash flows, and income statements are available for the last 4 annual periods or 4 quarters only. While sufficient for MVP validation, Phase 1 will migrate to professional feeds (Bloomberg/Refinitiv) to meet the "Institutional Readiness" standards described in Section 5.2.
    - Rate limiting: Approximately 2,000 requests per hour per IP address can be made. Exceeding this triggers temporary HTTP 429 blocks.
    - Unofficial API: yfinance and yahoo-fin scrape Yahoo's public endpoints and can break if Yahoo changes website structure.
    - Due to captcha issues, workflows are delayed by purposely adding sleep commands to act as a human (almost 10-15 seconds).
- Finviz:

    - Locked Features: Finviz explicitly gates capabilities behind Elite (premium paid tier). For e.g. 50 portfolios in free vs 100 in elite; 3 years statement history vs 8 years.
    - Scraping limiting: Finviz's bots explicitly disallows crawling key endpoints.
- AlphaVantage (Backup Layer):
    - It is constrained by the free tier limit of 25 requests per day.
    - The system logic accounts for this by prioritizing FinViz and Yahoo, engaging AlphaVantage only when strictly necessary to fill critical data gaps.
- AI Model Limitations:
    - Forward-Looking Bias: Off-the-shelf models (like GPT-4) are trained on vast, indiscriminate datasets that include future market knowledge relative to our backtesting periods. As noted in academic literature (Kargarzadeh, 2022), this can artificially inflate performance metrics and provide a forward-looking bias.
    - Data Leakage & Privacy: Transmitting investment theses to external API providers introduces potential data leakage risks, which is incompatible with strict banking secrecy standards.
- Miscellaneous:
    - Fundamentals are snapshots: Metrics like P/E ratio, beta, and market cap reflect current values only; historical trends for these ratios cannot be retrieved.
    - 15-minute price delay: Both Yahoo Finance and Finviz have a delay of at least 15 minutes i.e. Data is not real-time; suitable for backtesting and analysis but not for live trading decisions.

PatentsView (US Patent Database):

- US filings only: Does not capture patents filed in Europe (EPO), China (CNIPA), or other jurisdictions, potentially missing significant R&D activity for multinational companies.
- Assignee name matching: Addressed via a Legal Entity Resolution Layer that maps tickers to official corporate names (e.g., 'KO' → 'The Coca-Cola Company') combined with Fuzzy Text Search (_text_any) in the patent database to capture subsidiary filings.Subsidiaries, acquisitions, legal entity changes can cause incomplete results.
- API rate limits: The public API has undocumented rate limits; large portfolio runs may need throttling or require upgrading to premium access.

Assumptions used in the MVP

- Ticker symbols are valid and listed on supported exchanges (primarily US markets).

- All data is currently processed in US Dollars (USD), as the MVP focuses exclusively on the US market.
- Equal-weighted tickers are considered in the portfolio (for N number of tickers).
- The Yahoo Finance, Alpha Vantage and FinViz and PatentsView APIs are accessible and respond within the specified timeout windows.
- Date ranges typically fall within available coverage (2000-present for daily equity prices) or 25 tickers per day.
- Company names resolved from ticker symbols match PatentsView assignee records with a high degree of accuracy.
- Users understand patent data reflects US filings only and may not capture international innovation activity.
- Analysis is for informational purposes; real-time trading decisions require dedicated market data feeds.